

STATISTICS FUNDAMENTALS

A 10-DAY PRACTICAL GUIDE TO
CORE STATISTICAL CONCEPTS
AND DATA ANALYSIS



DESCRIBE
Understand Your Data



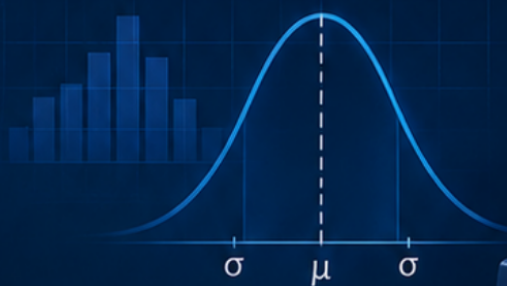
ANALYZE
Find Meaning



INTERPRET
Extract Insights



DECIDE
Drive Better Outcomes



**MASTER THE BASICS.
APPLY WITH CONFIDENCE.**

- ✓ Descriptive Statistics
- ✓ Data Distribution
- ✓ Outlier Detection
- ✓ Correlation Analysis
- ✓ Probability & More
- ✓ Real-World Applications

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Day 1 — Data Types & Classification

Definition

Data classification is the process of identifying whether variables are qualitative or quantitative and determining subtype.

Core Elements

- Qualitative vs Quantitative
- Nominal vs Ordinal
- Discrete vs Continuous

Step-by-Step Solution Process

1. Review each variable.
2. Determine whether variable is categorical or numeric.
3. If numeric, classify as discrete or continuous.
4. Document classification and reasoning.

Excel Formulas / Workflow

- No Excel formula required; classification is conceptual.

Manual / Non-Excel Formulas

- Use logical definitions to categorize each variable.

Day 2 — Central Tendency

Definition

Measures of central tendency summarize the typical value of a dataset.

Core Elements

- Mean
- Median
- Mode

Step-by-Step Solution Process

1. Inspect dataset for skew/outliers.
2. Calculate mean.
3. Calculate median.
4. Calculate mode.
5. Compare mean vs median.

Excel Formulas / Workflow

- =AVERAGE(range)
- =MEDIAN(range)
- =MODE.SNGL(range)

Manual / Non-Excel Formulas

- Mean = Sum/Count
- Median = Middle Ordered Value
- Mode = Most Frequent Value

Day 3 — Variability

Definition

Measures of variability quantify spread in a dataset.

Core Elements

- Range
- Variance
- Standard Deviation

Step-by-Step Solution Process

1. Calculate range.
2. Compute sample variance.
3. Compute sample standard deviation.
4. Interpret spread.

Excel Formulas / Workflow

- =MAX(range)-MIN(range)
- =VAR.S(range)
- =STDEV.S(range)

Manual / Non-Excel Formulas

- Range = Max – Min
- Variance = $\Sigma(x - \bar{x})^2 / (n - 1)$
- Std Dev = $\sqrt{\text{Variance}}$

Day 4 — Quartiles & Percentiles

Definition

Quartiles and percentiles position values within a ranked distribution.

Core Elements

- Q1
- Median
- Q3
- Percentiles
- IQR

Step-by-Step Solution Process

1. Calculate quartiles.
2. Compute percentiles.
3. Calculate IQR.
4. Interpret ranking.

Excel Formulas / Workflow

- =QUARTILE.INC(range,1)
- =QUARTILE.INC(range,3)
- =PERCENTILE.INC(range,k)

Manual / Non-Excel Formulas

- $IQR = Q3 - Q1$

Day 5 — Distribution & Skewness

Definition

Distribution analysis examines how values are spread and whether they are skewed.

Core Elements

- Histogram
- Positive Skew
- Negative Skew
- Symmetric Distribution

Step-by-Step Solution Process

1. Calculate mean and median.
2. Create histogram.
3. Compare mean vs median.
4. Determine skew direction.

Excel Formulas / Workflow

- =AVERAGE(range)
- =MEDIAN(range)
- Insert → Histogram

Manual / Non-Excel Formulas

- Positive Skew if $\text{Mean} > \text{Median}$
- Negative Skew if $\text{Mean} < \text{Median}$

Day 6 — Outlier Detection

Definition

Outliers are unusually high or low observations.

Core Elements

- IQR Method
- Lower Bound
- Upper Bound

Step-by-Step Solution Process

1. Calculate Q1 and Q3.
2. Compute IQR.
3. Calculate bounds.
4. Flag outliers.

Excel Formulas / Workflow

- =QUARTILE.INC(range,1)
- =QUARTILE.INC(range,3)
- =Q1-(1.5*IQR)
- =Q3+(1.5*IQR)

Manual / Non-Excel Formulas

- Lower = $Q1 - 1.5(IQR)$
- Upper = $Q3 + 1.5(IQR)$

Day 7 — Correlation

Definition

Correlation measures strength and direction of relationship between two variables.

Core Elements

- Pearson Correlation
- Scatter Plot
- R^2

Step-by-Step Solution Process

1. Ensure both variables are numeric.
2. Check for outliers.
3. Create scatter plot.
4. Calculate correlation.
5. Interpret strength/direction.

Excel Formulas / Workflow

- =CORREL(x_range,y_range)
- Insert → Scatter Plot

Manual / Non-Excel Formulas

- $r = \frac{\sum[(x - \bar{x})(y - \bar{y})]}{\sqrt{(\sum(x - \bar{x})^2 \times \sum(y - \bar{y})^2)}}$

Day 8 — Data Visualization

Definition

Data visualization communicates patterns and comparisons graphically.

Core Elements

- Line Chart
- Bar Chart
- Histogram
- Scatter Plot

Step-by-Step Solution Process

1. Define business question.
2. Choose chart type.
3. Prepare data.
4. Build chart.
5. Extract insight.

Excel Formulas / Workflow

- Insert → Appropriate Chart Type

Manual / Non-Excel Formulas

- Trend → Line
- Comparison → Bar
- Distribution → Histogram
- Relationship → Scatter

Day 9 — Probability

Definition

Probability measures the likelihood of an event occurring.

Core Elements

- Simple Probability
- Conditional Probability

Step-by-Step Solution Process

1. Count favorable outcomes.
2. Count total outcomes.
3. Compute simple probability.
4. Restrict denominator for conditional probability.

Excel Formulas / Workflow

- =COUNTIF()/COUNTA()
- =COUNTIFS()/COUNTIF()

Manual / Non-Excel Formulas

- $P(A) = \text{Favorable} / \text{Total}$
- $P(A|B) = P(A \cap B) / P(B)$

Day 10 — Mini Statistical Project

Definition

Exploratory statistical analysis combines multiple techniques to derive business insights.

Core Elements

- Descriptive Stats
- Outliers
- Correlation
- Pivot Tables
- Visualizations

Step-by-Step Solution Process

1. Profile dataset.
2. Compute descriptive stats.
3. Run outlier checks.
4. Test correlations.
5. Create pivots and visuals.
6. Synthesize insights.

Excel Formulas / Workflow

- =AVERAGE()
- =MEDIAN()
- =STDEV.S()
- =CORREL()
- Pivot Tables

Manual / Non-Excel Formulas

- Apply all prior statistical formulas as needed.